

Homework # 5: Bayesian hypothesis testing and credible sets

Due February 28

1. Let $\mathbf{X}$ be a sample of size n from $\text{Uniform}(0, \theta)$ distribution, where θ has $\text{Pareto}(\alpha, \sigma)$ prior (exactly as in Homework #4). Construct a $(1 - \alpha)100\%$ HPD credible set for the parameter θ .
2. A large shipment of parts is received, out of which five are tested for defects. The number of defects X has $\text{Binomial}(5, \theta)$ distribution. From the history of past shipments, it is known that θ has a $\text{Beta}(1, 9)$ prior distribution. Suppose that $X = 0$ is observed.
 - (a) Find the 95% HPD credible set for θ .
 - (b) Is there a significant evidence that the proportion of defective parts in the whole shipment exceeds 0.1? Conduct a Bayesian test comparing posterior probabilities of the null hypothesis $H_0 : \theta \leq 0.1$ and the alternative hypothesis $H_A : \theta > 0.1$.
 - (c) Revise your conclusion to the previous question taking into account that accepting a large shipment with more than 10% of defective parts is very costly, and thus, the Type II error results in a loss that is 5 times the loss due to the Type I error.
 - (d) Compute the Bayes factor for the test in (b,c).
3. (Berger 4.28) From path perturbation of a nearby sun, the mass θ of a neutron star is to be determined. Five perturbations quantified as 1.2, 1.6, 1.3, 1.4, and 1.4 are observed. Each observation is (independently) normally distributed with mean θ and assumed standard deviation $\sigma = 0.2$. A priori nothing is known about θ , so the noninformative prior density $\pi(\theta) \equiv 1$ is used. Find a 90% HPD credible set for θ .
4. Review, practice, prepare for the midterm.